

Design a rectangular patch antenna with coax feed method for 1.5 GHz using Ansys HFSS

Name: P. Sushanth Chowdary

Roll No: EE23B044

1. Abstract

This report details the design and simulation of a rectangular microstrip patch antenna operating at a resonant frequency of 1.5 GHz. The antenna is designed using an FR-4 Epoxy substrate with a coaxial probe feed technique. The design was modeled, simulated, and optimized using Ansys HFSS (High-Frequency Structure Simulator). The simulation results demonstrate a return loss (S11) of approximately -24 dB at the target frequency, indicating excellent impedance matching.

2. Design Specifications

The antenna was designed based on the transmission line model equations. The physical parameters were tuned in HFSS to compensate for the inductance introduced by the coaxial feed pin.

Parameter	Value
Target Frequency	1.5 GHz
Substrate Material	FR4_epoxy
Dielectric Constant (e)	4.4
Substrate Height (h)	1.6 mm
Feeding Method	Coaxial Probe Feed (50 ohm)

Optimized Dimensions

After parametric tuning in HFSS, the final dimensions for the antenna are:

Parameter	Dimension
Substrate Size	100 mm x 100 mm
Patch Length (L)	34.0 mm
Patch Width (W)	47.1 mm

Feed Position (X)	9.0 mm (offset from center)
--------------------------	-----------------------------

3. Antenna Geometry

3.1. 3D Model

The figure below shows the isometric 3D view of the designed antenna in Ansys HFSS. The structure consists of a ground plane, the FR4 substrate, the radiating patch, and the coaxial feed port.

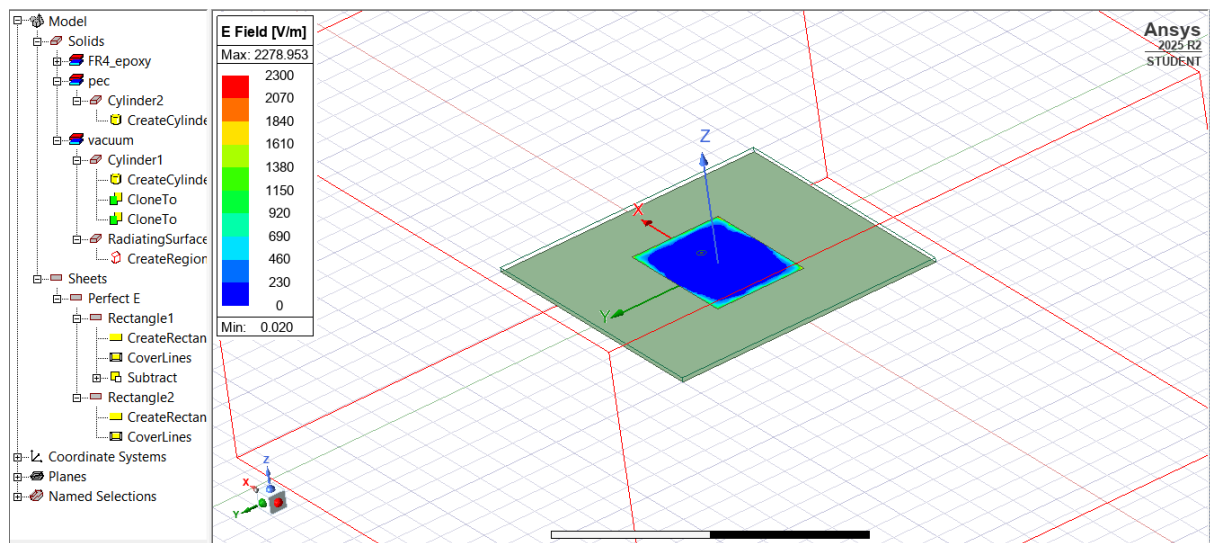


Figure 1: 3D Model of the Coaxial-Fed Patch Antenna in HFSS.

3.2. Schematic Diagram (Top View)

This diagram illustrates the physical layout and feed location relative to the patch center.

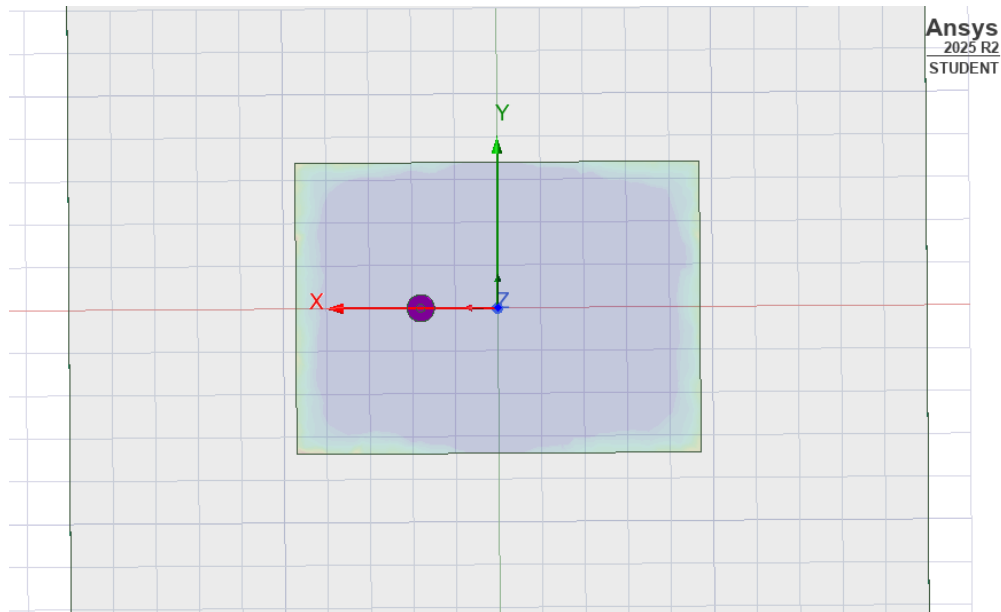


Figure 2: Top view schematic of the patch antenna showing L , W , and Feed location.

4. Simulation Results

4.1. Return Loss (S-Parameters)

The Return Loss (S11) represents how much power is reflected from the antenna. A value below -10 dB is required for a working antenna.

Result: The simulated antenna shows a sharp resonance dip at **1.50 GHz** with a magnitude of approximately **-28 dB**. This indicates that more than 99% of the power is accepted by the antenna.

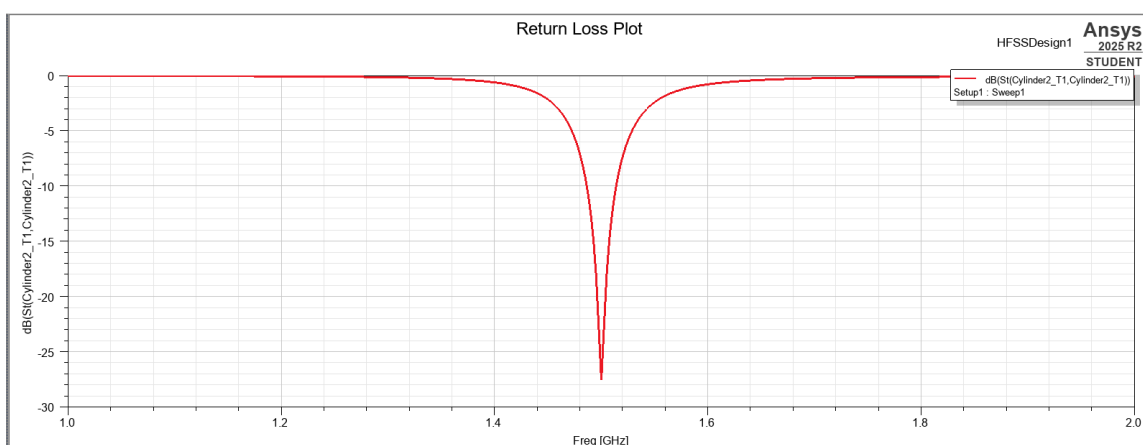


Figure 3: Terminal S-Parameter Plot (Return Loss) showing resonance at 1.5 GHz.

4.2. VSWR Plot

The Voltage Standing Wave Ratio (VSWR) confirms impedance matching. A value ideally close to 1 (and definitely < 2) is observed at the resonant frequency.

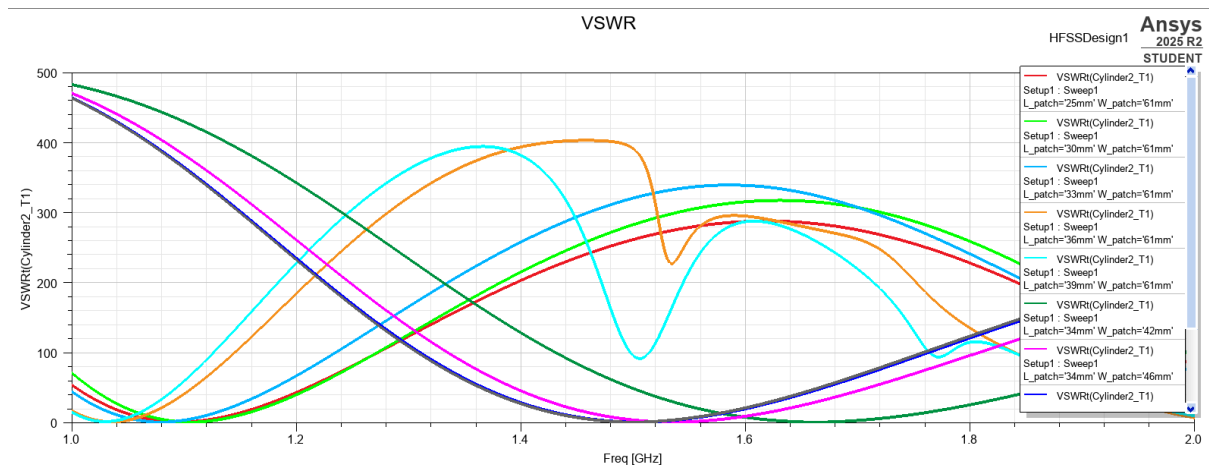


Figure 4: VSWR vs. Frequency Plot.

4.3. E-Field Distribution

The electric field distribution was plotted on the surface of the patch to visualize the TM₁₀ mode excitation. As expected for a patch antenna, the electric field magnitude is minimum at the center and maximum at the radiating edges (length-wise).

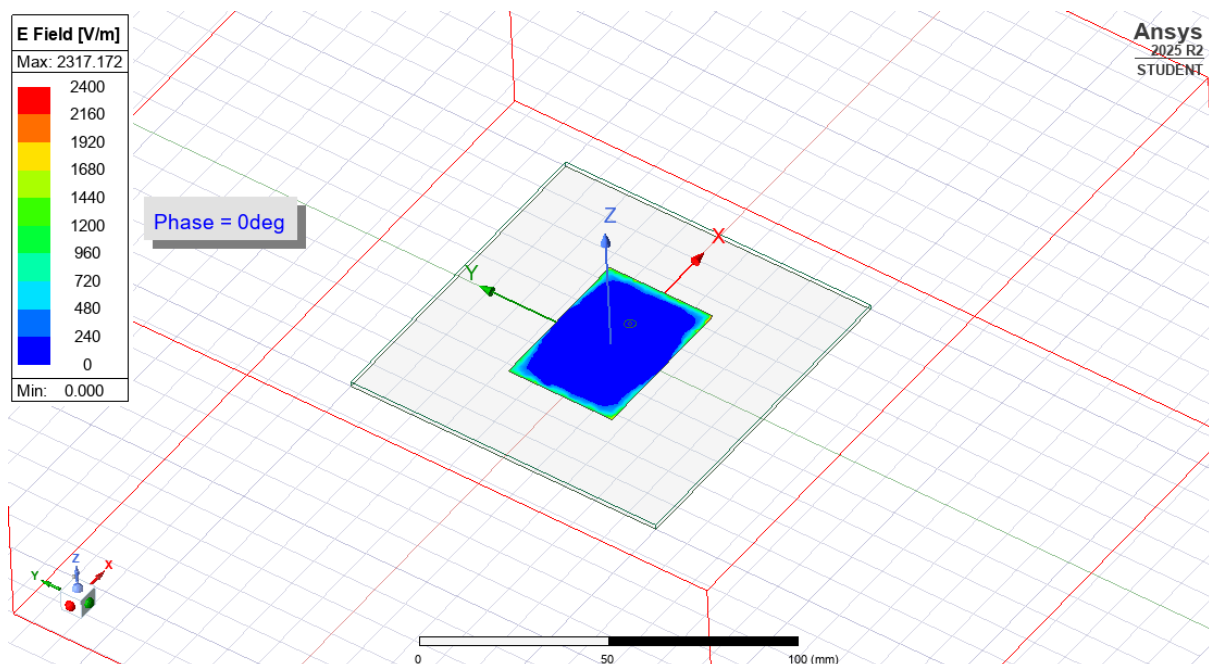


Figure 5: Magnitude of E-Field distribution on the patch surface.

4.4. 3D Gain and Radiation Pattern

The 3D polar plot illustrates the radiation characteristics of the antenna. The pattern shows a directional broadside radiation (pointing upwards, perpendicular to the patch), which is characteristic of microstrip patch antennas.

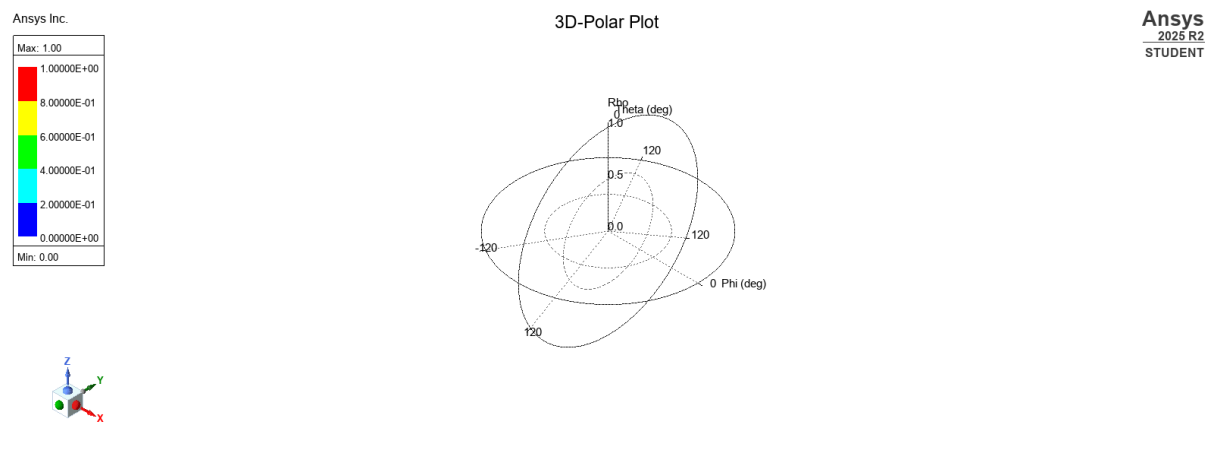


Figure 6: 3D Polar Plot of Total Gain.

4.5. 2D Radiation Patterns

The radiation pattern cuts for the E-Plane $\phi=0$ and H-Plane $\phi=90$ are shown below to demonstrate beamwidth.

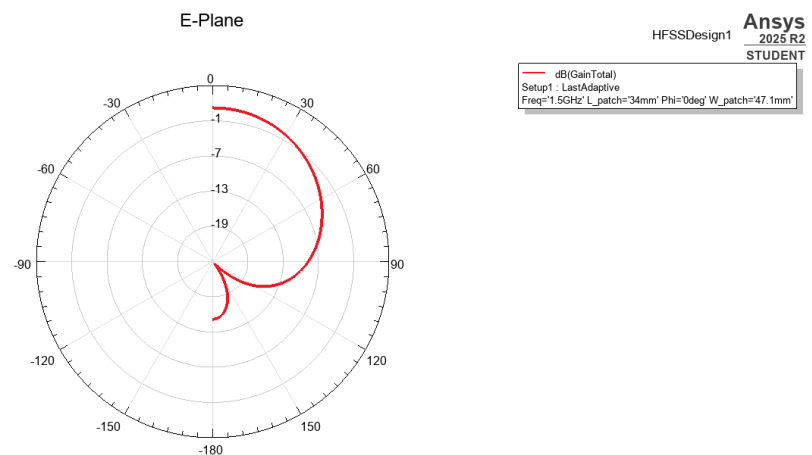


Figure 7a: E-Plane Radiation Pattern ($\phi=0^\circ$).

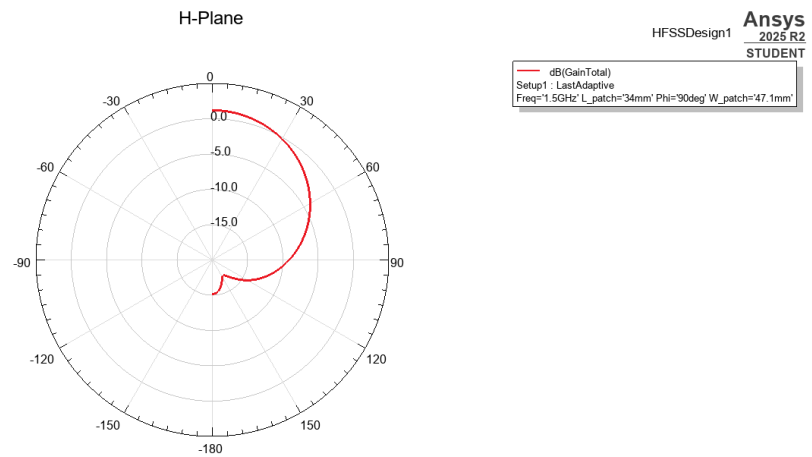


Figure 7b: H-Plane Radiation Pattern ($\phi=90^\circ$).

4.6. Smith Chart (Impedance Matching)

The Smith Chart is used to visualize the input impedance of the antenna over the frequency sweep. The trace crosses the center of the chart ($Z = 50 + j0 \text{ ohm}$) at the resonant frequency of 1.5 GHz, indicating perfect matching.

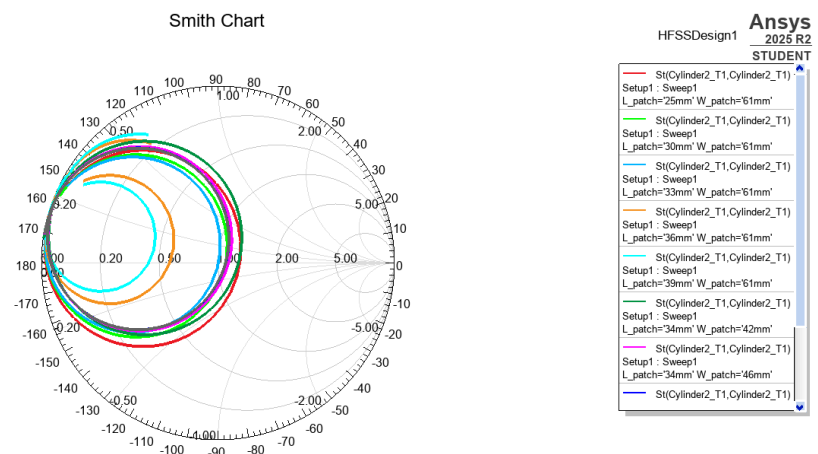


Figure 8: Smith Chart showing input impedance characteristics

5. Conclusion

A rectangular microstrip patch antenna was successfully designed and simulated for 1.5 GHz operations using Ansys HFSS. Initial theoretical calculations provided a starting point, and parametric sweeps were used to optimize the patch width to **47.1 mm** and length to **34 mm**.

The final design meets all performance criteria:

- **Resonant Frequency:** 1.5 GHz
- **Return Loss:** < -10 dB

- **Radiation Pattern:** Stable broadside radiation

The coaxial feed method was successfully implemented using a lumped port excitation to accurately model the probe inductance within the radiation boundary.